

First Proposal.docx

KING FAHD UNIVERSITY OF PETROLEUM & MINERALS Electrical Engineering Department Senior Design Project EE 411 Term # 172 Section #		
Project Proposal Title.....		
	Name:	ID#
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Instructor:		

Date: DD/MM/YYYY

Project Proposal

Project definition:

Implementing a complete wireless seismic data acquisition prototype and designing some parts of the system.

In this project, we will develop an electronics system that can be connected to a generic geophone sensor to process, digitize and sample the collected data.

Project Objectives:

- Win Capstone Design competition
- Utilize the accumulated knowledge and experience gained from the university courses.
- To gain deep understanding by dealing with practical real-life applications.
- Accomplish working prototype.
- Utilize the gained experience for future industrial participation.
- Contribute to the field of seismic data acquisition.
- Learn modern technologies.
- Improve the team work skills.

Specifications:

- Wireless distance range to be around 50 meters.
- Weight does not exceed 1.5 kilograms.
- SNR at least 50dB.
- Dimension of the wireless prototype less than 20cm x 20cm x 10cm.
- Utilize more than 80% of the ADC dynamic range.
- Cost of the wireless device should not exceed 5
- Power
- Modulation type
- Working frequencies:
 - o M&F
 - o Bla
 - o Blo
 - o Blee

Quantization levels (at least 10 bits resolution)

portable power supply with long life.

Filter's bandwidth.

SNR.

Constraints:

- Cost should not exceed 2000 SR
- Time limitation of only 14 weeks
- Technology availability (for example we can not fabricate the designed circuit.
- Measurement equipment limited capability.
- Components availability.
- Lack of deep knowledge and practical experience.

Standards:

Sensor

Pre-amplifier and filter

IEEE 1241-2010 (Standards for Terminology and Test Methods for Analog-to-Digital Converters).

Data acquisition

IEEE 802.15.4 (low-rate wireless personal area networks standards)

IEEE 754-2008 (Binary Floating Point Arithmetic Standard).

IEEE Std C95.1™-2005 IEEE Standard for Safety Levels with Respect to Human Exposure to Radio Frequency Electromagnetic Fields, 3 kHz to 300 GHz

Strategy:

Revise the prerequisite courses.

Regular Group Meetings.

Divide the work among team members

Searching and analyzing several existing designs.

Construct system blocks (bottom up top down?)

Testing and troubleshooting of electrical circuits and code algorithms.

Implementing the optimum design that achieves the desired specifications.

Plan of action:

Proof of concept

Understanding the idea of the seismic data acquisition systems.

Choose an appropriate design.

Collect the needed components.

Implement the design.

Verify the results and make the necessary adjustment.

Design Component: Identify the design part in your project.

Main Circuits:

- Sensor
 - o Coil based geophone
 - o Gyroscope
 - o Bezos electric
 - o MEMS (micro electromechanically system)
- Preamplifier circuit
- ADC circuit

- Data acquisition
 - o Arduino
 - o FPGA
 - o MyRio
- Wireless communication port (such as Arduino RF transceiver module)
- Body Case

Project Outcomes: Spell out the expected outcomes in a bullet fashion describing each outcome in a short sentence.

- Master preamplifiers
- Complete working prototype
- Detailed design report
- Gather solid knowledge in electronics field.
- Emphasize on priority management skills

Budget: Do cost analysis of your project counting the man-hours needed and the cost of the material.

- Components
-
- Documents
- Software

Man Power	Material
20 hour/week = 4 hours/week/person	genaric geophone
	microcontroller
	ADC
	Pre-Amplifier

Reporting: Weekly progress report, early presentation, final presentation and final report

Not Yet

Week1: Mojtaba

Week2: Mohannad

Week3: Faisal S

Week4: Abdullah

Week5: Faisel M

Prototype: At the end of the project, a working prototype will be presented.

Not Yet

Tasks/Time schedule:

Final Report												X	X	X
Presentation			x	x										
Project Proposal		x	x											
Group Meetings	x	x	x	x	x	x	x	x	x	x	x	x	x	x

Tasks/Time schedule: